

Sujoy Das

Senior Software Engineer | Distributed Data Platforms, ML Infrastructure & Cloud Systems

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SUMMARY

Senior Software Engineer with 10+ years building distributed data and ML platforms on Spark, Iceberg, and Kafka, benchmarked to **25B+ time-series readings across 3M devices**. End-to-end ownership across contract-driven ingestion, row-level data quality, multi-region release automation, model serving, and run-level observability. Led delivery of four analytics products, directed six engineers, and drove source-contract alignment between Data Science and consumer teams.

TECHNICAL SKILLS

Languages & Testing: Python, Java, Scala, SQL, REST APIs, gRPC, Microservices, JUnit, Mockito, Pandas

Data & Pipelines: Spark, PySpark, Databricks, Structured Streaming, Kafka, Airflow, ETL, ELT, Incremental Loads

Cloud & CI/CD: OCI, AWS, Kubernetes, Docker, Terraform, Infrastructure as Code, CI/CD, GitOps, GitLab CI/CD, Jenkins

Storage & Formats: Apache Iceberg, Delta Lake, Parquet, Lakehouse, Data Lake, Data Warehouse, Oracle ADW

Data Quality & Modeling: Schema Evolution, Data Modeling, Idempotent Writes, Data Lineage, Governance, Pandra

Observability & Performance: Structured Logging, Metrics, Prometheus, Jaeger, Kibana, Distributed Systems, Fault Tolerance

ML & GenAI: TensorFlow, ONNX, XGBoost, Model Registry, Model Serving, RAG, LLM Integration, Vector Search

PROFESSIONAL EXPERIENCE

Oracle | Senior Software Engineer

Sep 2024 – Present

- Engineered a distributed multi-tenant Spark ingestion framework benchmarked at **25B+ time-series readings** from 3M devices and now running **13,000+ source files daily** across two commodity types, scaling to absorb a **2x volume increase**.
- Led delivery of four Meter Operations analytics products** by onboarding four engineers, aligning Data Science and MDM teams on source contracts, and designing a governed **Iceberg-to-Kafka** lakehouse publishing layer that merged three Spark outputs into one business record to allow safe schema evolution.
- Leading author** of the **multi-region** release and orchestration engine for OCI Data Science jobs, validating region, tenant and onboard-versus-upgrade semantics in the merge request, and versioning model artifacts with their Conda environments and release metadata in Git so every job traces to its build and commit.
- Cut root-cause time on Spark performance issues by **80%** by engineering **Spark Analyzer**, a Python diagnostics engine that parses raw event logs into linked DAG models and applies 12 deterministic rules including skew, spill and GC pressure, now the team standard.
- Removed a shared failure domain** in a distributed system where one failed branch stalled every dependent job by moving fan-out upstream to a component that already tracked each file's interval and unit, turning serial coupling into fault-tolerant, **independently routable parallel branches** with lineage preserved.

Oracle | Software Engineer

May 2021 – Aug 2024

- Cut compute by **50%+** against the batch baseline at equivalent throughput by redesigning a 5-stage batch flow to near-real-time Structured Streaming, removing end-of-batch latency and mentoring the engineer through the build.
- Eliminated code changes** for new dataset onboarding by engineering a declarative validation engine driven by JSON contracts, executing per-partition **Pandra** checks through **mapInPandas** and retaining all failure reasons so bad files stay explainable and runs replay reliably.
- Productionalized ML model serving on **two distinct inference paths, XGBoost-to-ONNX** on PySpark with grouped Pandas UDFs over schema-validated features, and batched **TensorFlow** in OCI Data Science Model Catalog.
- Standardized **observability** across **all pipeline projects**, cutting support triage time by engineering a centralized metrics layer that enriches every execution stage with tenant, job, and operational context to enable direct cross-run analysis.

HealthEdge | Associate Developer

Aug 2019 – Apr 2021

- Improved core Java claims-processing throughput by **more than 50%** across all product versions by adding a caching layer and tuning query execution plans and indexes.
- Cut regression validation from **24 hours to 4** by partitioning the test dataset across Docker worker containers, each verifying its own slice and reporting pass or fail to a coordinator, and raised unit test coverage from 30% to 85%.

Mphasis Ltd | Delivery & Associate Software Engineer

Apr 2014 – Jun 2017

- Led a **team of 6 engineers** delivering J2EE and Spring backend services and REST APIs, owning Jenkins CI/CD and JMeter performance testing with zero unplanned downtime.

EDUCATION

Master of Science, Computer Science | Binghamton University, State University of New York

Aug 2017 – May 2019

B.E., Computer Engineering | Rashtrasant Tukadoji Maharaj Nagpur University, India

Aug 2009 – May 2013

CERTIFICATIONS: OCI Generative AI Professional | Oracle AI Data Platform Essentials | OCI Foundations Associate